

Section: Our Dynamic Universe

Topic: Gravitation

A spacecraft of mass $1.2 \times 10^4 \text{ kg}$ is in orbit around a planet of mass $7.4 \times 10^{22} \text{ kg}$ and radius $1.6 \times 10^6 \text{ m}$. The spacecraft is in a circular orbit at a height of $2.4 \times 10^5 \text{ m}$ above the planet’s surface.

(a)
Calculate the gravitational field strength at this height

Use the formula:

$$g = \frac{GM}{r^2}$$

Where:

- $G = 6.67 \times 10^{-11} \text{ Nm}^2\text{kg}^{-2}$
- $M = 7.4 \times 10^{22} \text{ kg}$
- $r = 1.6 \times 10^6 + 2.4 \times 10^5 = 1.84 \times 10^6 \text{ m}$

$$g = \frac{6.67 \times 10^{-11} \times 7.4 \times 10^{22}}{(1.84 \times 10^6)^2} = \frac{4.9358 \times 10^{12}}{3.3856 \times 10^{12}} \approx 1.46 \text{ N kg}^{-1}$$

(b)
Calculate the orbital speed of the spacecraft

Use:

$$v = \sqrt{\frac{GM}{r}} = \sqrt{\frac{6.67 \times 10^{-11} \times 7.4 \times 10^{22}}{1.84 \times 10^6}}$$

$$v = \sqrt{\frac{4.9358 \times 10^{12}}{1.84 \times 10^6}} = \sqrt{2.6825 \times 10^6} = 1.64 \times 10^3 \text{ m s}^{-1}$$

 Revision Tips

- Always check whether the question gives **radius from centre** or **height above surface** — if height is given, **add the planet’s radius**
- The formula for g outside a planet is:

$$g = \frac{GM}{r^2}$$

- Orbital speed comes from equating gravitational force to centripetal force:

$$v = \sqrt{\frac{GM}{r}}$$

- Use **standard form** consistently and double-check exponents